

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

III B. Tech I Semester Regular Examinations, November - 2025

**FORMAL LANGUAGES AND AUTOMATA THEORY
(CSE)**

Time: 3 hours

Max. Marks: 70

Note:

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Questions	BTL	CO	Marks
1.	a) State the applications of Automata Theory.	K1	CO1	2M
	b) Explain the differences between Mealy and Moore machine models.	K2	CO1	2M
	c) Define Arden's Theorem.	K1	CO2	2M
	d) Design the regular expression that represents all possible strings consisting only of the symbol 0 or 1.	K2	CO2	2M
	e) Define ambiguous grammar and illustrate with an example.	K2	CO3	2M
	f) State the Pumping Lemma for context-free languages.	K1	CO3	2M
	g) Explain when a PDA accepts a string.	K2	CO4	2M
	h) Describe unit productions and null (ϵ) productions.	K2	CO4	2M
	i) List the special features of a Turing Machine.	K1	CO5	2M
	j) Define the Church-Turing Hypothesis.	K1	CO5	2M

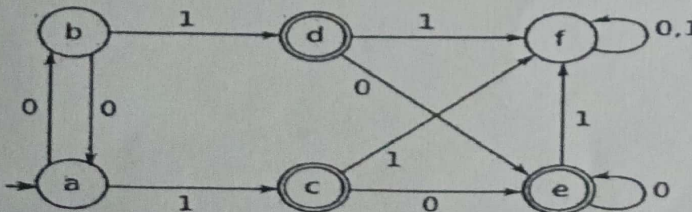
UNIT – I

2.	a)	Construct a DFA that recognizes odd number of 1's and even number of 0's.	K3	CO1	5M
	b)	Construct the equivalent DFA for the given NFA. (q_0 – Initial state and q_2 – Final state)	K3	CO1	5M

δ/Σ	a	b
$\rightarrow q_0$	{ q_0, q_1 }	{ q_2 }
q_1	{ q_0 }	{ q_1 }
$* q_2$	$\emptyset$	{ q_0, q_1 }

OR

3.	a)	Describe the standard steps for minimizing a DFA and explain how each step works.	K2	CO1	5M
	b)	Explain the procedure to minimize the given finite automaton, listing each step and the reason for it.	K2	CO1	5M



UNIT – II

4.	a)	Construct a finite automaton equivalent to the regular expression $(0+1)^*$	K3	CO2	5M
	b)	State the Pumping Lemma for regular languages and provide its proof.			
OR			K4	CO2	5M

5.	a)	Construct a regular expression that generates all strings consisting of alternating 0's and 1's, and justify how the expression captures this pattern.	K3	CO2	5M
	b)	Differentiate between right and left linear regular grammars			
UNIT – III			K2	CO2	5M

6.	a)	Define derivation tree? Construct Leftmost and Rightmost derivation for the string 11001010. $S \rightarrow 1B/0A$ $A \rightarrow 1/1S/0AA$ $B \rightarrow 0/0S/1BB$	K3	CO3	5M
	b)	Convert the following grammar into Greibach normal form. $S \rightarrow AA/a$ $A \rightarrow SS/b$			
OR					
			K3	CO3	5M

7.	a)	Eliminate all unit productions from the grammar $S \rightarrow AB$ $A \rightarrow E$ $B \rightarrow C$ $C \rightarrow D$ $D \rightarrow b$ $E \rightarrow a$ and present the resulting grammar without unit productions.	K3	CO3	5M
	b)	What is Left factoring? Apply left factoring to the following grammar $A \rightarrow abB / aB / cdg / cdeB / cdfB$	K3	CO3	5M

8.	a)	Explain the concept of an instantaneous description (ID) in a pushdown automaton and describe how it represents the current computation state.	K2	CO4	5
	b)	Construct a pushdown automaton that recognizes the language generated by the grammar $S \rightarrow aB$ $B \rightarrow bA/b$ $A \rightarrow aB$	K3	CO4	5

9.	a)	Explain the differences between deterministic pushdown automata and nondeterministic pushdown automata.	K2	CO4	5M
	b)	Construct a pushdown automaton (PDA) that accepts all strings over {0,1} containing an equal number of 0's and 1's.	K3	CO4	5M

10.	a)	Explain the concept of a Universal Turing Machine and describe its significance.	K2	CO5	5M
	b)	Construct a Turing Machine that accepts the language $L = \{a^n b^n, n > 0\}$ and present an instantaneous description (ID) for the input string 'aabb'.	K3	CO5	5M

11.	a)	Describe tractable and intractable problems and outline their key characteristics.	K2	CO5	5M
	b)	Explain the differences between the Post Correspondence Problem and the Modified Post Correspondence Problem.			